

Algebra 1: Polynomials

Challenge Level

Name: _____

Date: _____

Question 1: Solve the equation $x^2 + 5x - 6 = 0$

- A) $x = -2, x = 3$
- B) $x = -3, x = 2$
- C) $x = -6, x = 1$
- D) $x = -1, x = 6$
- E) $x = 2, x = -3$

Question 2: Factor the polynomial $x^2 + 7x + 12$

- A) $(x + 3)(x + 4)$
- B) $(x + 2)(x + 5)$
- C) $(x + 1)(x + 6)$
- D) $(x - 3)(x - 4)$
- E) $(x - 2)(x - 5)$

Question 3: Simplify the expression $2x^2 + 5x - 3 + 3x^2 - 2x - 1$

- A) $5x^2 + 3x - 4$
- B) $5x^2 + 2x - 2$
- C) $5x^2 + 3x - 2$
- D) $x^2 + 3x - 4$
- E) $x^2 + 2x - 2$

Question 4: Find the value of x for which $x^2 - 4x - 5 = 0$

- A) $x = -1, x = 5$
- B) $x = 1, x = -5$
- C) $x = -5, x = 1$
- D) $x = 5, x = 1$
- E) $x = -1, x = -5$

Question 5: Solve the inequality $x^2 - 2x - 6 > 0$

- A) $x < -2, x > 3$
- B) $x > -2, x < 3$
- C) $x < 2, x > -3$
- D) $x > 2, x < -3$
- E) $x > -6, x < 1$

Question 6: Find the equation of the axis of symmetry for the parabola $x^2 + 4x + 4$

- A) $x = -2$
- B) $x = 2$
- C) $x = 1$
- D) $x = -1$
- E) $x = 0$

Question 7: Solve the equation $x^2 + 2x - 15 = 0$

- A) $x = -5, x = 3$
- B) $x = -3, x = 5$
- C) $x = 3, x = -5$
- D) $x = 5, x = 3$
- E) $x = -3, x = -5$

Question 8: Find the value of x for which $x^2 - 9x + 20 = 0$

- A) $x = 4, x = 5$
- B) $x = 5, x = 4$
- C) $x = 4, x = -5$
- D) $x = -5, x = 4$
- E) $x = 5, x = -4$

Question 9: Prove the factor theorem, which states that $x - c$ is a factor of $f(x)$ if and only if $f(c) = 0$. Provide examples to support your proof.

Question 10: Find the roots of the cubic equation $x^3 - 6x^2 + 11x - 6 = 0$ by factoring. Provide the factored form of the equation and the values of the roots.

Chef's Tips (Hints):

- To solve polynomial equations, first try factoring the left-hand side of the equation.
- When simplifying expressions, combine like terms to make the expression easier to work with.
- To find the roots of a cubic equation, try factoring it or use the rational root theorem to find possible rational roots.